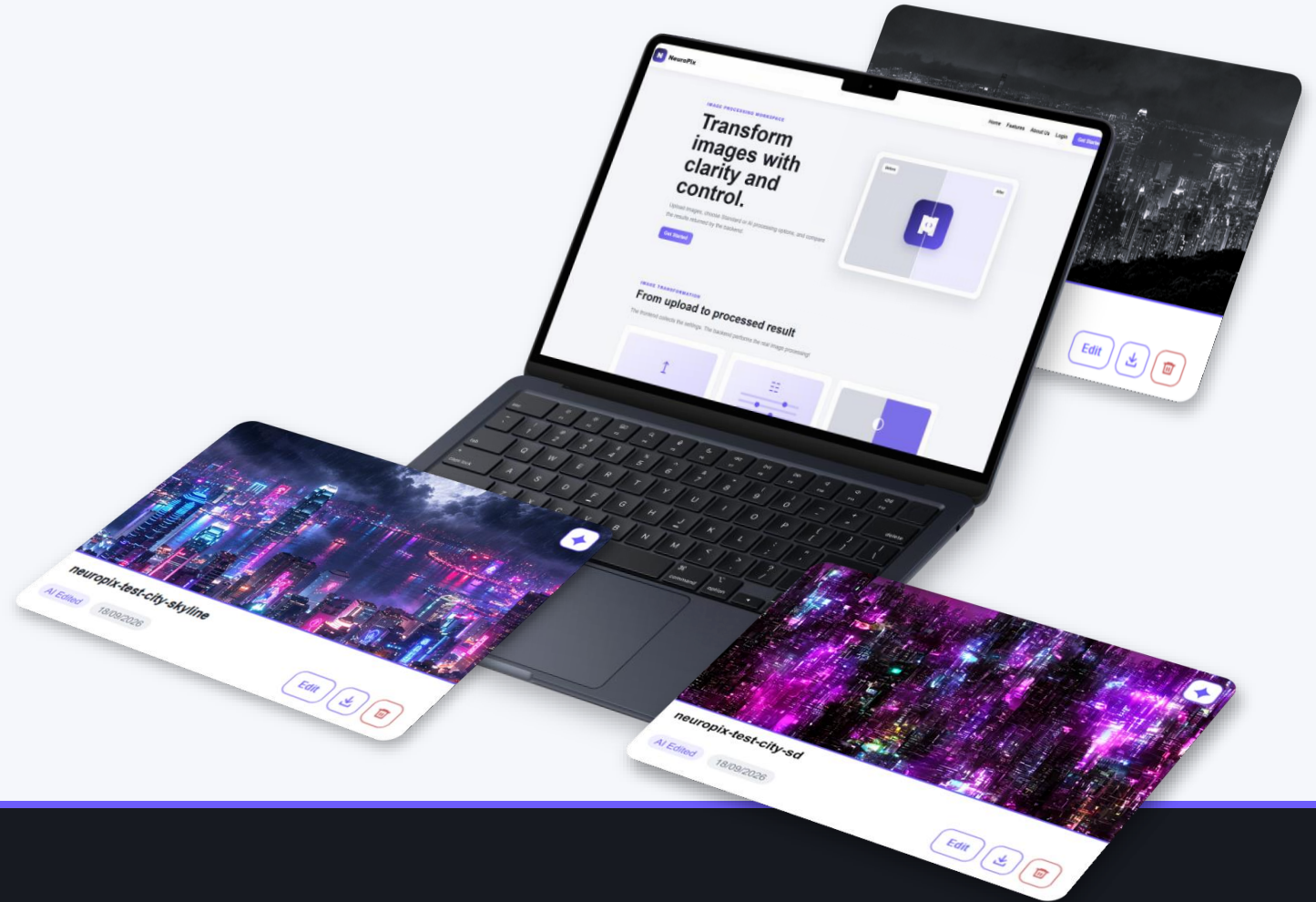


NeuroPix

Image editing and AI
in one browser workspace

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Supervisor: Dr. Faisal Arafsha

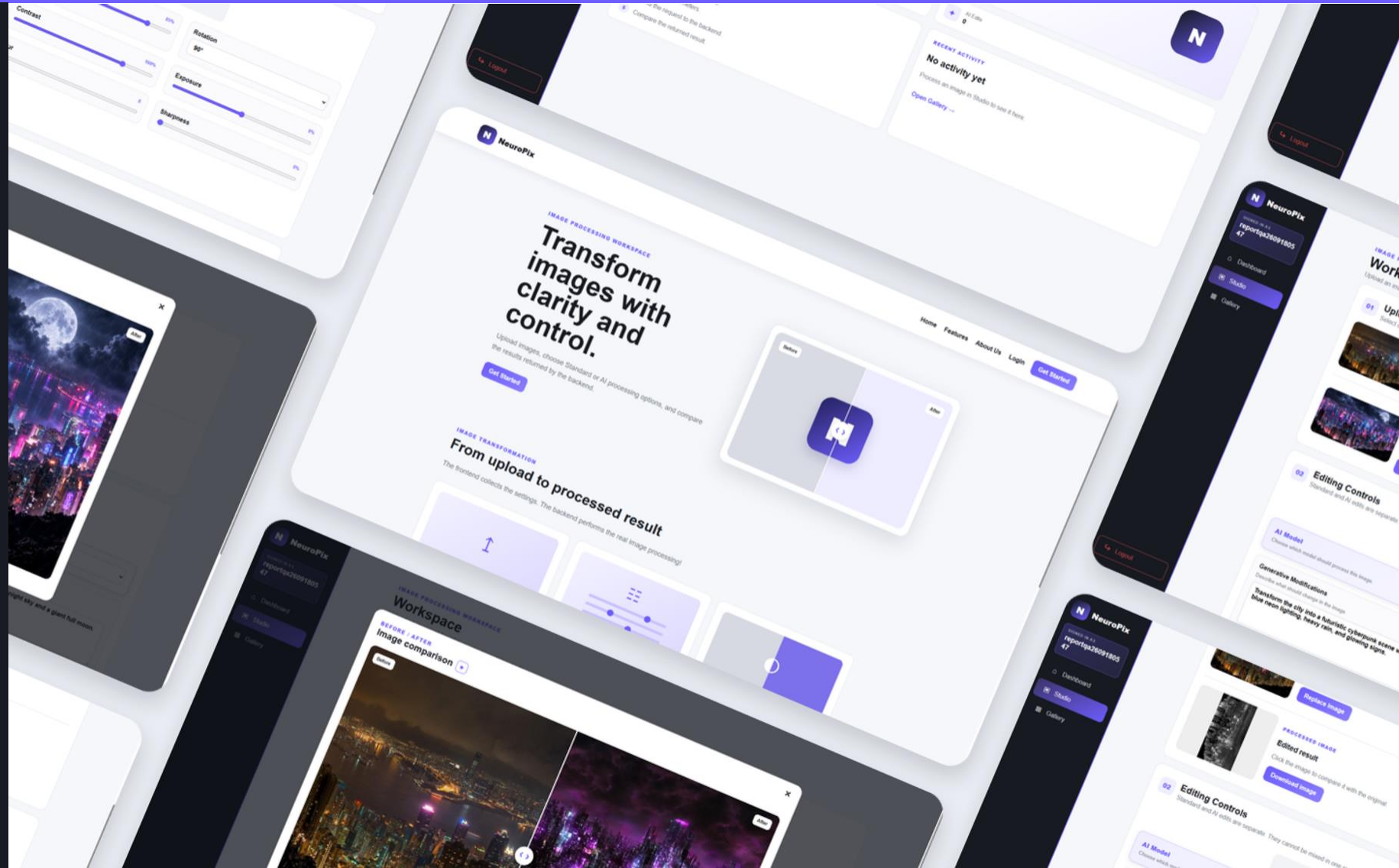
BSc Software Engineering · 2026

Image Editing in One Workspace

Standard and AI image editing in your browser.

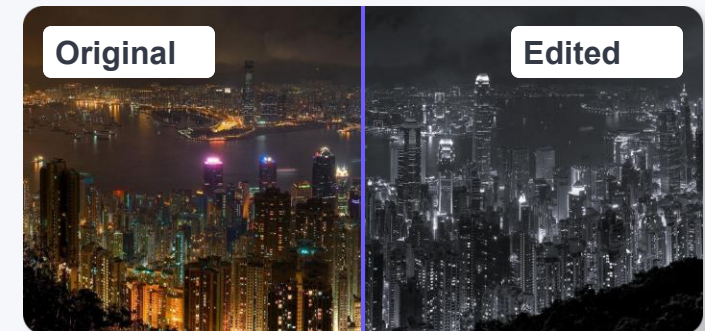
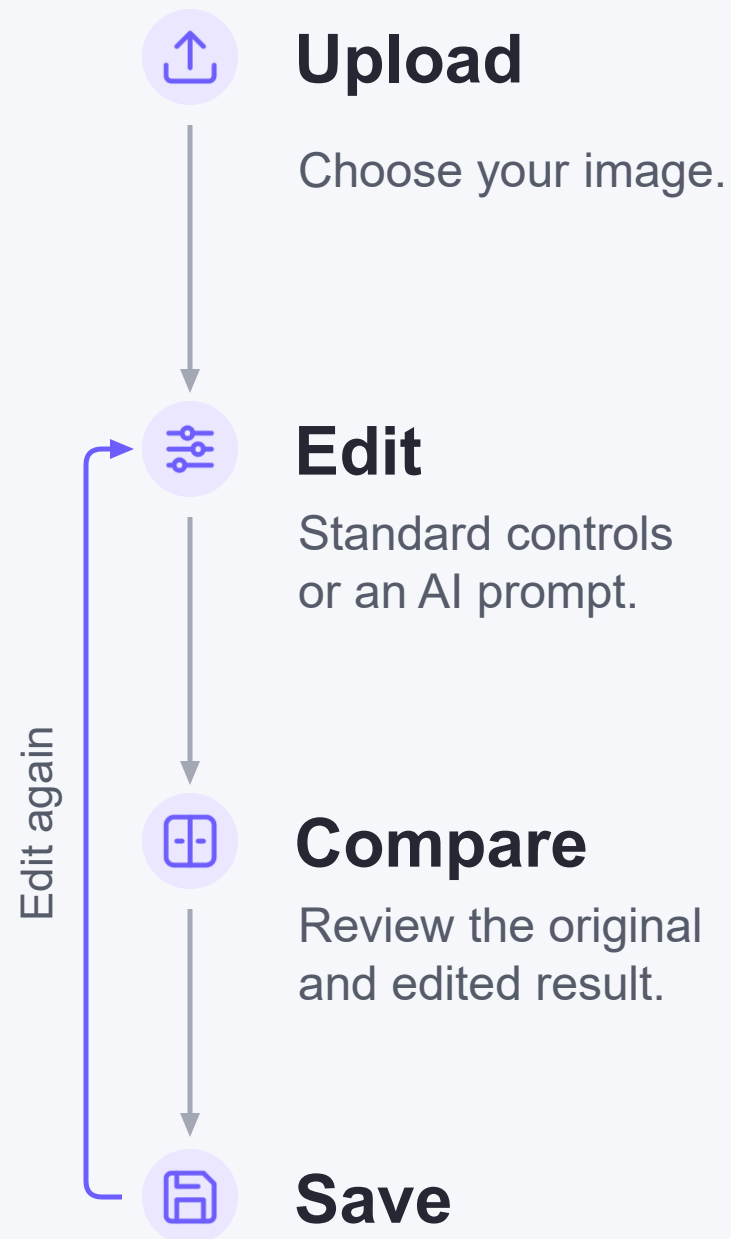
**Simplicity and
easy access**

Use standard controls
or describe an AI edit,
then compare the
result before saving.



The Editing Workflow

Upload once.
Refine the same
image.



Gallery or download

Project Development

From browser editor to a deployed, tested service.



01 · FOUNDATION

Build the editor

Put standard image controls
in a browser workspace.

BROWSER · STANDARD EDITS



02 · INTEGRATION

Connect the pieces

Connect OpenAI and local AI.
Save images and metadata.

OPENAI · LOCAL SD · S3 / RDS



03 · DELIVERY

Deploy and test

Expose the site over HTTPS,
then verify the live use cases.

READY FOR THE LIVE DEMO

Standard Editing

Several adjustments, one edited result.

Selected adjustments

Grayscale

100%



Contrast

160%



Brightness

115%



AI Editing Prompts

The edit fields are handled differently by each provider.

02 Editing Controls

Standard and AI edits are separate. They cannot be mixed in one request.

Standard Edits

AI Edits

AI Model

Choose which model should process this image.

OpenAI

Generative Modifications

Describe what should change in the image.

Transform the city into a futuristic cyberpunk scene with intense purple and blue neon lighting, heavy rain, and glowing signs.

Enhancement

Describe the quality or detail improvement.

Add crisp reflections, dramatic contrast, and cinematic detail.

Background Manipulation

Describe the background you want.

Replace the daylight sky with a dark stormy night sky and a giant full moon.

Upscaling

Choose an enlargement level.

1x



OpenAI

A pre-prompt adds rules for each edit field.

EDIT FIELDS



PRE-PROMPT

S. Stable Diffusion

The local route combines the selected edit text into one prompt.

EDIT FIELDS



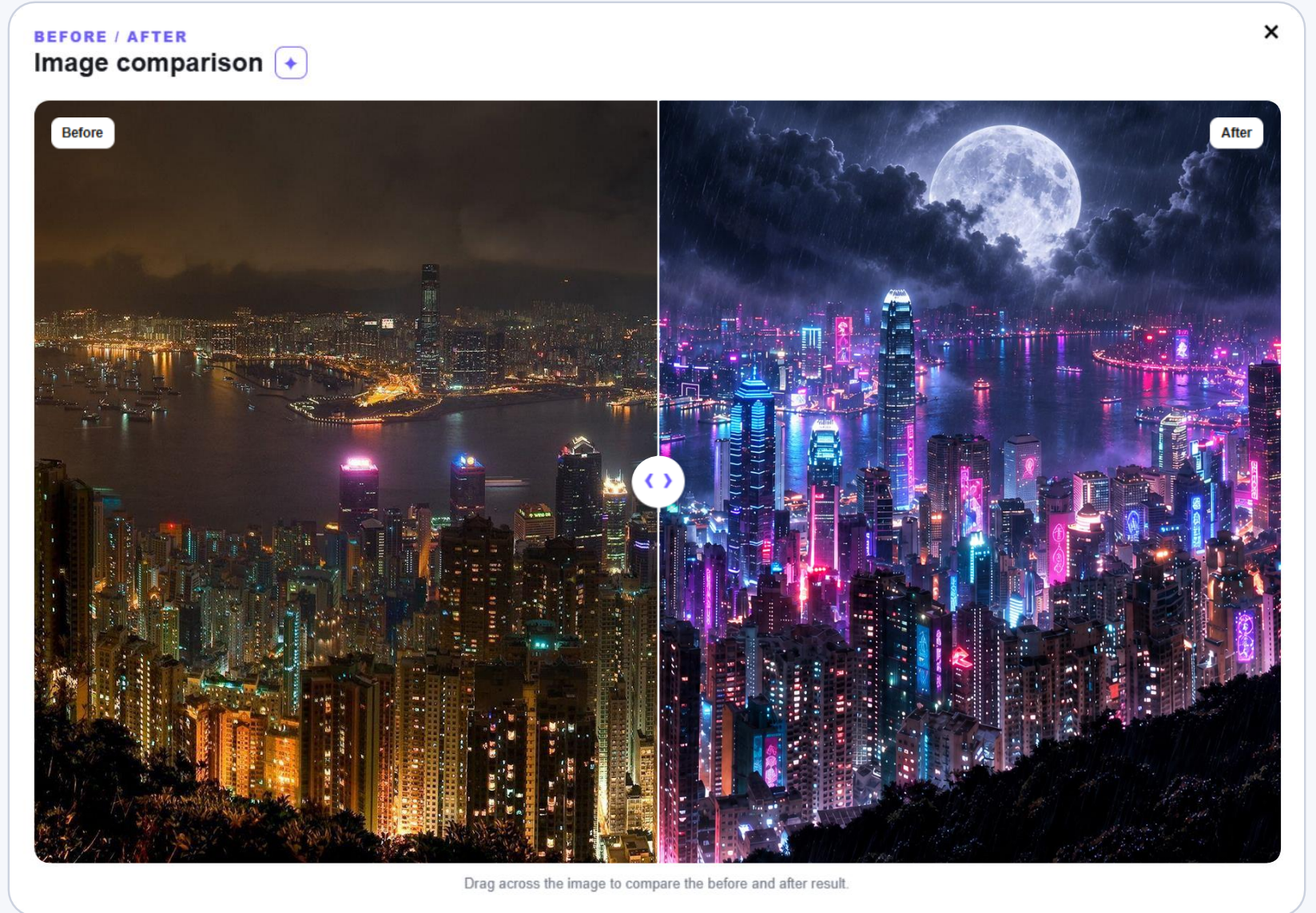
ONE PROMPT

OpenAI Image Editing

Cyberpunk Cityscape

A moonlit sky and neon lighting reshape the original skyline.

DEFAULT PROVIDER



BEFORE / AFTER

Image comparison 

×



Drag across the image to compare the before and after result.

Local Stable Diffusion

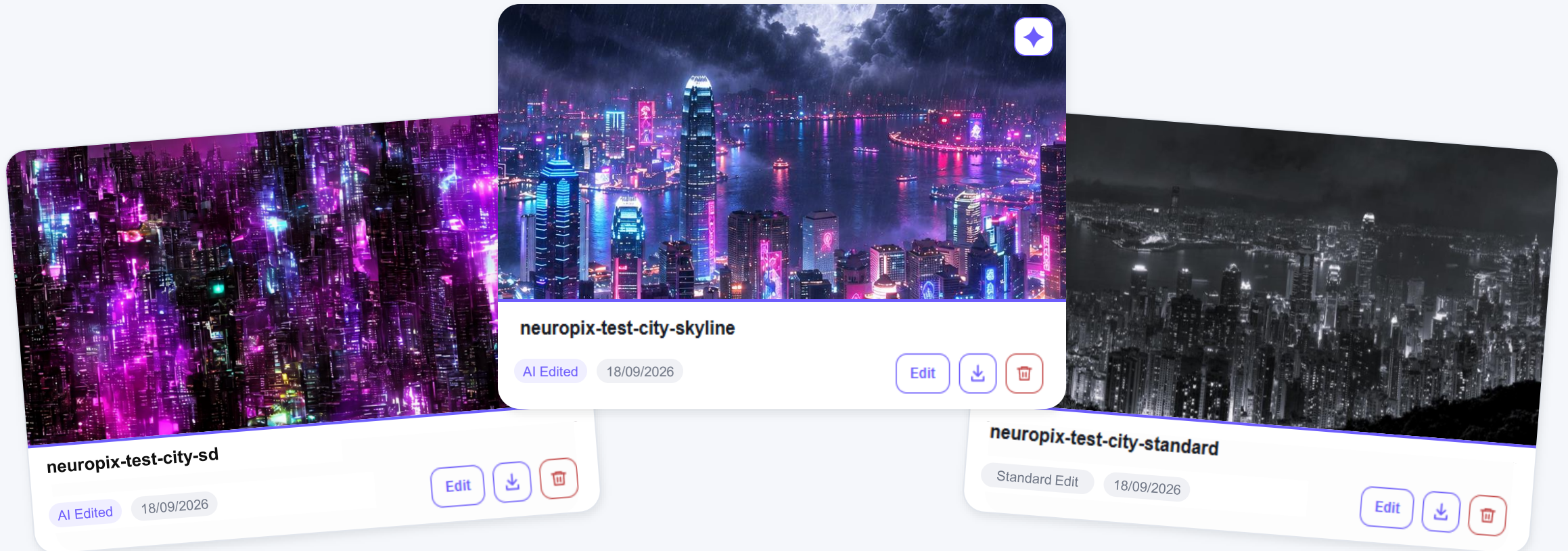
Neon Cityscape

The edit adds purple clouds and neon lights to the original skyline.

SELF-HOSTED MODEL

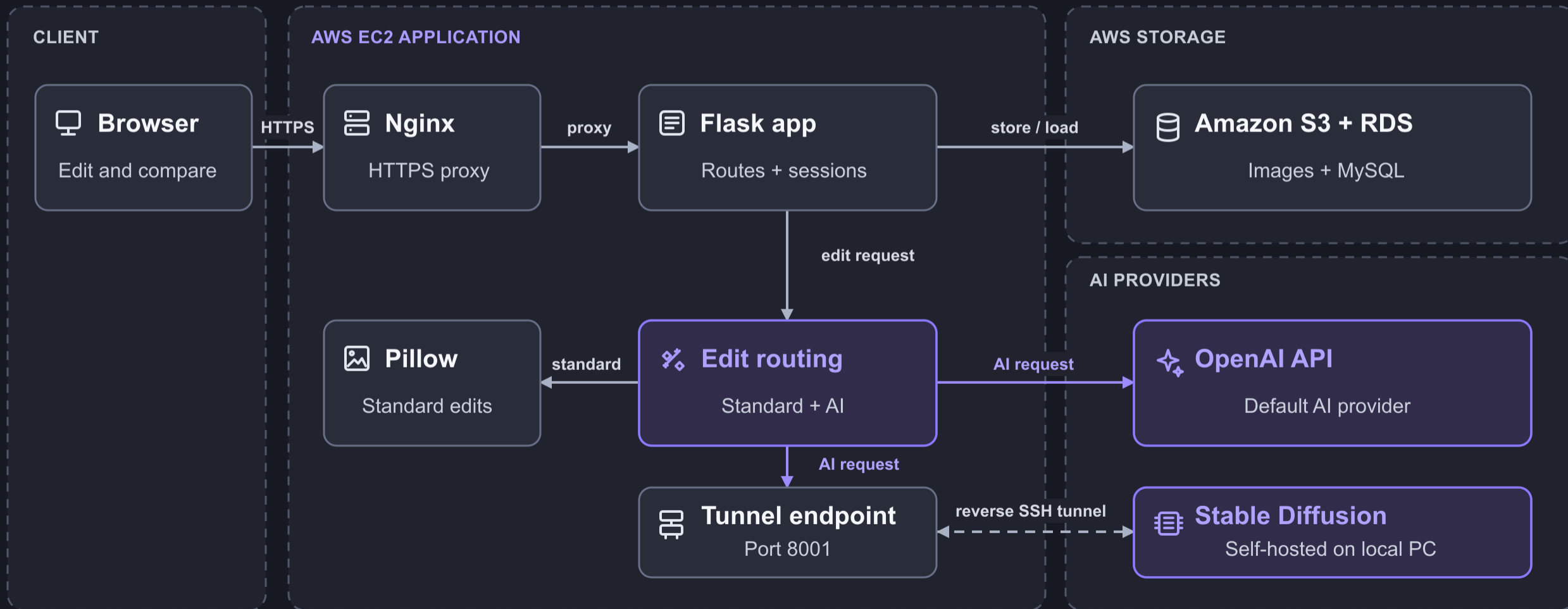
Saved Images

Revisit, download, or delete saved images.



Application Architecture

The EC2 application handles standard edits and connects to AWS storage and AI providers.



Development Choices

Cost and ease of learning guided our choices.



DOMAIN

Namecheap

The free student offer covered our domain.



HOSTING

AWS

Popular services gave us learning resources and hands-on cloud deployment experience.



TEAMWORK

GitHub

Working on main kept collaboration simple while teammates learned Git.

AI Model Choices

A hosted API for easy integration, a local model to control costs.



OpenAI

GPT-Image-2.5 Sunburst

Easy API integration
New image model available
during development



S. Local Stable Diffusion

Version 1.5

Fits our RTX 3080
Avoids hosted GPU costs

Six Live Use-Case Tests

Live testing on the deployed website.

ACCESS

02

 Register

 Sign in

EDITING

03

 Standard edit

 OpenAI edit

 Stable Diffusion
edit

GALLERY

01

 Find, download, and delete

Test Results

Automated checks and live website testing.

AUTOMATED TESTS

35/35

 **Passed**

API, security, and pages

LIVE USE-CASE TESTS

6/6

 **Passed**

Authentication, editing, gallery

Future Improvements

The next improvements to reliability and everyday use.



Storage recovery

Improve recovery when image storage or database updates fail.



Processing status

Show processing status while an edit is running.



Browser compatibility

Test editing in Firefox, Edge, and mobile browsers.



Next: Live Demo

<https://neuropix.me>

NEUROPIX · SENIOR PROJECT



Thank You

Questions & Discussion

<https://neuropix.me>